MIL-M-45521B(WC)
30 November 1971
SUPERSEDING
MIL-M-45521A(WC)
26 April 1965

MILITARY SPECIFICATION

MULTIMETER: M17

1. SCOPE

1.1 This specification covers one type of portable, self-contained, precision multimeter used to conduct safety and reliability tests on the 762mm Rocket (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-V-173	-	Varnish, Moisture and Fungus Resistant (For Treatment of Communications, Electronic and Associated Equipment).
MIL-T-4807	-	Tests, Vibration and Shock, Ground Electronic Equipment (General Requirements for).
MIL-S-6872	-	Soldering Process, General Specification for.

STANDARDS

Federal

FED-STD-595	-	Colors.

Military

MIL-STD-109	-	Quality Assurance Terms and Definitions.
MIL-STD-130	-	Identification Marking of U.S. Military Property.
MIL-STD-171	-	Finishing of Metal and Wood Surfaces.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-810	-	Environmental Test Methods.

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DRAWINGS

U.S. Army Weapons Command

D8100900	-	Multimeter, M17.

(Part drawings referenced in this specification form a part of Drawing D8100900.)

PUBLICATIONS

U.S. Army Weapons Command

IEL 8100900	-	Index of Inspection Equipment List.
P8100900	-	Packaging Data Sheet.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Welding Society

AWS-A2.0	Welding Symbols

(Application for copies should be addressed to the American Welding Society Inc., 345 East 47th Street, New York, New York 10017.)

3. REQUIREMENTS

3.1 First article. Requirements for submission of first article shall be as specified in the contract (see 6.2). Unless otherwise specified, the first article shall include the pilot pack (see 5.1).

3.2 Materials and construction. Multimeters shall conform to the materials and construction requirements specified herein and on Drawing D8100900 and drawings applicable thereto.

3.2.1 Welding. Welding of aluminum alloy and steel parts shall be as specified on the drawings and in accordance with the applicable welding specification. Actual production of weldments shall not proceed prior to Government approval of the manufacturer's welding procedures and test specimens. Interpretation of drawing weld symbols shall be in accordance with AWS-A2.0.

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3.2.2 Soldering. Soldering of all electrical components shall be in accordance with MIL-STD-454, Requirement 5. Soldering of nonelectrical components shall be in accordance with MIL-S-6872.

3.2.3 Protective finishes.

3.2.3.1 Inorganic finishes. Inorganic finishes, including metal deposited coatings and surface treatment, shall be in accordance with the applicable finish number of MIL-STD-171 as specified on the drawings.

3.2.3.2 Organic finishes. Painting shall be in accordance with the applicable finish number of MIL-STD-171 as specified on the drawings.

3.2.3.3 Moisture and fungus resistant treatment. Moisture and fungus resistant treatment shall be in accordance with MIL-V-173 as specified on the drawings.

3.3 Design. Multimeters shall conform to the design requirements specified herein, on Drawing D8100900 and drawings applicable thereto.

3.3.1 Vibration. The multimeter shall be capable of withstanding, without breakage, deformation, loosening of parts or electrical damage, the vibrations normally encountered during transportation and handling of equipment (see 4.5.3.6).

3.3.2 Shock. The multimeter shall be capable of withstanding, without breakage, deformation, loosening of parts or electrical damage, the shocks normally encountered during transportation and handling of equipment (see 4.5.3.7).

3.3.3 Calibration. The assembled multimeter shall be calibrated in accordance with the procedure specified on Drawing F8100953. After calibration, potentiometers shall be locked. A small seal (sealing wax) shall be applied to the potentiometer stem and locking nut. An unbroken seal indicates that calibration has not been disturbed.

3.4 Marking. Marking of the multimeter, including identification of parts and instructional marking, shall be as specified on the applicable drawings and in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Quality assurance terms and definitions. Quality assurance terms and definitions used herein are in accordance with MIL-STD-109.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. Inspection of first article (see 3.1) and pilot pack (see 5.1) shall be performed at the designated testing agency. First article and pilot pack shall be subjected to the inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract (see 6.2).

4.5 Quality conformance inspection.

4.5.1 Inspection lot. All multimeters produced under essentially the same condition shall be considered a lot for inspection purposes.

4.5.2 Examination.

4.5.2.1 Multimeter. Each multimeter shall be visually examined for cleanliness; workmanship; incorrect assembly or missing parts; illegible, incorrect or missing markings and instructions; visible defects of painting including incomplete coverage, wrinkles, sags, chipped surfaces or wrong color. Manually examine for incorrect function of cover catches. Multimeters having one or more defects shall be rejected.

4.5.2.2 Packaging and packing. Examination of packing and packaging of multimeters shall be in accordance with Packaging Data Sheet P8100900.

4.5.3 Test methods.

4.5.3.1 Material testing. Tests required by applicable drawings and specifications shall be performed on materials used in the manufacture of multimeter assemblies. The contractor shall furnish the Government representative certified statement of findings of chemical analyses and reports of physical tests as may be required for each lot or each material used. The contractor shall also furnish the Government representative certification that the materials used in packaging conform to the applicable specifications.

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4.5.3.2 Welding. Inspection and testing of weldments and qualification of welders shall be in accordance with the welding specifications referenced on the applicable drawings.

4.5.3.3 Soldering. Inspection and testing of soldering shall be in accordance with the soldering specifications referenced on the applicable drawings.

4.5.3.4 Protective finishes. Protective finishes of the multimeter, including inorganic coatings and painting, shall be in accordance with the applicable finish number of MIL-STD-171 as specified on the drawings.

4.5.3.5 Moisture and fungus resistant treatment. Inspection and test for moisture and fungus resistant treatment shall be in accordance with MIL-V-173.

4.5.3.6 Vibration test. Two multimeters drawn at random from each lot of 200 or less shall be tested in accordance with Method 1A of MIL-T-4807. If either fails to pass this test, acceptance testing shall be deferred pending corrective action by the contractor. After certification of corrective action, the test shall be repeated on two multimeters other than those previously tested. Failure of either multimeter to pass the test shall be cause for rejection of the lot.

4.5.3.7 Shock test. Two multimeters drawn at random from each lot of 200 or less shall be tested in accordance with Method 516, Procedure II or III of MIL-STD-810, with the exception that the peak value of the shock pulse shall be 20G. If either fails to pass this test, acceptance testing shall be deferred pending corrective action by the contractor. After verification of corrective action, the test shall be repeated on two more multimeters, other than those previously tested. Failure of either multimeter to pass the test shall be cause for rejection of the lot.

4.5.3.8 Calibration. Each multimeter shall be calibrated in accordance with Drawing F8100953 and shall be verified. The locking nut of each potentiometer shall be inspected manually and visually for compliance with locking and seal application requirements of 3.3.3. Failure to meet calibration requirements shall be cause for rejection.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. A pilot package shall be forwarded in accordance with the contract (see 6.2). Pilot packages shall be packaged to the level of packaging specified in the contract and packed in accordance with the requirements of Packaging Data Sheet P8100900.

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5.2 Preservation, packaging, packing and marking. Multimeters shall be preserved, packaged and marked in accordance with the requirements of packaging data sheet P8100900.

5.3 Repair parts. Repair parts shall be prepared for delivery with applicable packaging data sheets as specified in the contract (see 6.2).

6. NOTES

6.1 Intended use. The M17 multimeter is designed to make all necessary safety and reliability tests on the 762mm Rocket, M31 series and XM50 electrical systems. The tester is provisioned to check the continuity and insulation value of the main and spin motor squibs, the spin motor switch and the associated wiring. It is not intended to verify or check special circuits connected with warheads or fusing.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. List of drawings and publications pertinent to the M17 Multimeter assembly, showing applicable revision dates.
c. Requirements for submission of first article (see 3.1).
d. Requirements for submission of pilot pack (see 5.1).
e. Selection of applicable levels of preservation, packaging and packing.
f. Repair parts, when required.

6.3 Color. The applicable color chips of FED-STD-595 should be furnished the contractor for use as a guide for painting.

Custodian:
Army - WC

Review activities:
Army - SM

Preparing activity:
Army - WC

Project number:
1055-A033

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